STAT 578 B1: Advanced Causal Inference
Fall 2025

(Updated on Oct 26)

Lectures:	2055 Sidney Lu Mech Engr Bldg	Mon, Wed, Fri 2:00-2:50 pm
Instructor:	Chan Park	parkchan@illinois.edu 136 Computing Applications Building (CAB)
TA:	Rong Wei	rongw6@illinois.edu
Office Hours:	Available upon request	
Credits:	GR (4)	

Course Information

Course Overview

The course aims to equip graduate students with the foundational knowledge and skills needed to begin research in causal inference. Topics covered include: (i) key concepts in causal inference, such as potential outcomes, confounding, missing data, and causal graphs; (ii) identification, estimation, and inference of causal estimands; and (iii) recent advancements in causal inference, such as machine learning-based causal inference approaches. See the Tentative Class Schedule for details.

By the end of the course, students are expected to know or be able to do the following:

- Understand the mathematical language underpinning causal inference (potential outcomes) and be able to read technical papers on causal inference.
- Understand and explain key ideas in causal inference, including, but not limited to, ignorability/exchangeability/selection on observables, confounding, consistency/SUTVA, positivity, randomized experiments, instrumental variables (IV) and regression discontinuity designs, matching, propensity scores, inverse probability weighted (IPW) estimators, standardization/g-formula, double-robustness, treatment effect heterogeneity, semiparametric efficiency, interference, and sensitivity analysis
- Implement and apply key statistical methods in causal inference using R, including (i) matching, (ii) IPW, g-formula, and doubly robust methods with machine learning, (iii) IV methods, (iv) sensitivity analysis; and (v) etc.

Prerequisite

This is a graduate-level seminar course in statistics, primarily designed for Ph.D. students interested in learning about or conducting research in causal inference. The course places a strong emphasis

on methodology, computing, and data analysis. The effective prerequisites are:

- (Either STAT 510 or STAT 511) and (either STAT 527 or STAT 528);
- or department consent.
- The R programming language and R-markdown are **required** for completing the assignments.

Textbooks

There is no required textbook for the course, but here are some suggestions:

- Hernán, Robins (2020). *Causal Inference: What If*. Chapman & Hall/CRC. [Link]
- Ding (2024). *A first course in causal inference*. Chapman & Hall/CRC. [Link]
- Wager (2024). *Causal Inference: A Statistical Learning Approach*. Draft. [Link]
- Chernozhukov, Hansen, Kallus, Spindler, Syrgkanis (2024). *Applied Causal Inference Powered by ML and AI*. Draft [Link]
- Rosenbaum (2017). *Observation and experiment: An introduction to causal inference*. Harvard University Press.
- Imbens, Rubin (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press.
- Pearl (2009). *Causality*. Cambridge University Press.

Canvas

Canvas will be the hub for class material, grades, and announcements. All lecture slides, homework, etc. will be posted there. If you are new to Canvas, you can check this video.

Assignments and Submission Policies

- Homework:

  There will be approximately 3 homework assignments in total. For each lecture with an associated homework, you are required to write example R code implementing the topics covered, submitted in R Markdown (Rmd) format. Each homework assignment is due one week after the corresponding lecture. See the Tentative Class Schedule for the specific dates.

  Please sign up for 3 days (or more, if you wish) by visiting "Canvas → Calendar" by Sep 3, 2025. A maximum of three students can sign up per day, and scheduling will be on a first-come, first-served basis. If possible, please select days with no students assigned yet. The instructor and TA may request schedule adjustments to balance the number of students evenly across days. Students who do not sign up by the deadline will have their dates assigned by the instructor.

  Although homework assignments have been finalized, you may swap your assigned lecture order with classmates if unavoidable circumstances arise, provided that at least two students remain assigned to each lecture. If you make a swap after finalization, please notify both the instructor and the TA.

  You may use existing R packages; however, please clearly specify which packages are required for your implementation. You are encouraged to discuss your answers and code with classmates to review and improve your work. Nevertheless, each student must submit their own

complete Rmd file. If you incorporate ideas or assistance from classmates, please acknowledge their contributions appropriately. Failure to do so will be considered academic dishonesty and handled accordingly.

- Summary report of a presentation:

  Each of you is required to attend one presentation on causal inference (either in person or virtually) and write a summary report. The presentation that you attend must take place between Sep 1, 2025 and Dec 1, 2025. Please note that you may not summarize the instructor's presentation. Your report should be no longer than three pages, formatted with 11-point font, double spacing, single column, and 1-inch margins. Once you decide which presentation to attend, please share the details with the instructor to confirm that it meets the requirement. The report is due on Dec 14, 2025.

  Below are some suggested presentation options:

  - Midwest Econometrics Group Conference 2025 [Link]
    This event will be held on October 3-4 at UIUC. If you prefer in-person attendance, this is a great opportunity. Registration costs $50 and includes meals.
  - Statistics Department Seminar [Link]
    It is currently unclear whether a speaker specializing in causal inference will be featured, but if so, this would be a convenient opportunity to fulfill the assignment.
  - Online Causal Inference Seminar [Link]
    Various presentations on causal inference are available online.

- Paper presentation:

  At the end of the semester, you will give a 10-12 minute presentation on a paper. You may present your own ongoing or completed work on causal inference, a presentation you attended earlier in the semester, or recent works (ideally published in the last five years) from (bio)statistics journals (e.g., Journal of the American Statistical Association) or conference proceedings (e.g., NeurIPS). Please note that you may not present the instructor's work. Once you decide which paper to present, please share the details with the instructor to confirm that it meets the requirement. Please sign up by visiting "Canvas → Calendar" no later than Oct 31. There is a limit of students per day, and scheduling will be on a first-come, first-served basis.

Grading

Semester composite scores are calculated according to the breakdown shown below.

- Homework: $45\% = 3 \times 15\%$;
  If you submit more than 3 homework assignments, only your best 3 scores will be counted.

  Here is a rough rubric of each homework assessment:

  - 15 points: R code is comprehensive, including a clear summary of methods, explicit notes on package and dataset dependencies, interpretation of results, and runs without errors.
  - 10 points: Work shows good understanding but is missing some details or has minor errors.

  - 5 points: Significant issues in code or interpretation; many parts incomplete or incorrect.
  - 0 point: Homework was not submitted by the deadline.

  Since this is a graduate-level course designed to familiarize students with professional practices, your work may be reassessed through a revise-and-resubmit process similar to that used in journal or conference publications. If you are not satisfied with your homework score, you may revise and resubmit your code to the TA within three days of the score release. The TA will then re-evaluate the homework and adjust the score as appropriate. Requests for a higher score without substantive revision will not be considered.

- Summary report of a presentation: 25%

  Here is a rough rubric of the report:

  - 10 points: Summary of the presentation: May include:
    motivation for the presented work, contribution to the literature, key findings, simulation or data analysis results
  - 10 points: Evaluation of the work: May include:
    whether the claimed contributions and results are appropriately supported, or if they appear overstated; potential limitations of the work; whether the simulation or data application is well justified
  - 5 points: Evaluation of the presentation style (not the work itself): May include:
    aspects of the presentation you liked or disliked; suggestions for improving the delivery of key results; observations on how the presenter handled the Q&A session

- Paper presentation: 30%

  Here is a rough rubric of the presentation:

  - 15 points: Quality of the presentation: May include:
    whether the presentation provides an overall balanced summary of the work, effectively delivers the key ideas, and appropriately omits less important results.
  - 15 points: Evaluation of the presentation style: May include:
    whether the presentation is of the right length, has an appropriate level of technical detail, engages the audience sufficiently, and how the presenter handled the Q&A session.

Semester letter grades will be assigned based on your percentage of total points, broken down below:

- [90,100] $\in$ A- / A / A+
- [80,90) $\in$ B- / B / B+
- [70,80) $\in$ C- / C / C+
- [60,70) $\in$ D- / D / D+
- [0, 60) or missing either the report or presentation = F

Any rounding or curving will not be decided until after finals week when all grades are in. Please do not ask the instructor after each exam or before the end of the semester if there is any curve. I will have to refer you to this part of the syllabus.

Course Policies

Accommodations

To obtain disability-related academic adjustments and/or auxiliary aids, students with disabilities must contact the course instructor and the Disability Resources and Educational Services (DRES) as soon as possible. To contact DRES you may visit 1207 S. Oak St., Champaign, call 333-4603 (V/TTY), or e-mail a message to disability@illinois.edu. Note that homework and take-home exams are not eligible for DRES accommodations. Only the in-person exam qualifies for such accommodations.

Participation

Active participation in the learning environment is vital to your success in this course. Depending on your course, you may be asked to engage in discussions and other interactive learning environments that invite your active participation and involvement with other students and your instructor.

Student Commitment

By registering for this course, you commit to self-motivated study, participation in course activities, and timely submission of all assignments. Furthermore, you commit to accessing the course website and checking email at least twice per week, as well as to devoting at least 8–12 hours/week to prepare for each module and complete the required assignments and readings.

Responding to Emails (only for questions of personal nature)

Private emails to the TAs and the instructor should be sent for truly private matters. Private emails that ask questions about course materials, course administration, or homework questions may be sent, but replies are not guaranteed. Due to the large size of this course, we follow a strict email policy as detailed in course policies. In seeking help, please use office hours. For questions of personal nature, please visit the instructor or the course staff during office hours. If an online meeting is not possible given the personal situation, please state so in your private email to me or the course staff as appropriate.

Academic Dishonesty

Long story short: Do not engage in any form of academic dishonesty. The TAs and instructors are required to follow the university's policy and will take appropriate action if we find any evidence of it.

Academic dishonesty will not be tolerated. Examples of academic dishonesty include, but not limited to, the following:

- Cheating, Fabrication, Facilitating infractions of academic integrity, Plagiarism, Bribes, favors, and threats, Academic interference, Examination by proxy, Grade tampering, Non-original works, etc
- Please avoid sharing your answers or R code on any online platforms.

Should an incident arise in which a student is thought to have violated academic integrity, the student will be processed under the disciplinary policy set forth in the Illinois Academic Integrity

Policy: http://studentcode.illinois.edu/. Ignorance is not an excuse for any academic dishonesty. It is your responsibility to read this policy to avoid any misunderstanding. If you do not understand relevant definitions of academic infractions, contact your instructor for an explanation within the first week of class. The University of Illinois at Urbana-Champaign Student Code should also be considered as a part of this syllabus. Students should pay particular attention to Article 1, Part 4: Academic Integrity. Read the Code at the following URL: http://studentcode.illinois.edu/.

Collaboration, Copying, and Plagiarism

You are encouraged to discuss course material, including assignments, with your classmates. Homework assignments are meant to be learning experiences. You may discuss the exercises with other students, but you must write-up the solutions on your own. **In short, do not cheat, it is not worth the risk. You are more likely to get caught than you believe. If you think you may be operating in a gray area, you most likely are.** To ensure that you do not violate the academic integrity policies, all code and solutions should be prepared yourself. All code should be typed yourself, not copy and pasted. (Except for code provided in class)

All work you turn in must be your own. This includes both written explanations and code. Acknowledge any help received on your assignments. Copied assignments will receive no credit.

Copying from other students, books, websites, or solutions from previous versions of the class:

- does nothing to help you learn how to program,
- is easy for us to detect, and
- has serious negative consequences for you, as outlined in the university's policy on cheating and plagiarism which can be found in Article 1, Part 4 of the Student Code. Section 1-402 in particular outlines behavior which is considered an infraction of academic integrity. If, after reading the policy, you are unclear on what is acceptable, please ask the instructor.

The homework and exams obviously must be done completely independently. Copying or any other form of cheating will result in a zero score for the exam and possibly a more severe sanction, depending on the circumstances.

Likewise, posting any course material or asking answers on public sites like Coursehero/Discord/WhatsApp, etc, or making any course material available to other students is considered cheating and plagiarism. Please do not do it. Serious action will be taken.

Emergency Response Recommendations

If you have an emergency that will keep you from participating in the course, please notify your instructor by using the instructor's email address (not a Canvas message). Provide callback information in your email (if necessary). You should also notify your program director of any emergencies.

Emergency response recommendations can be found at the following website: https://police.illinois.edu/em/planning/emergency-response-guide/. I encourage you to review this website and the campus building floor plans website within the first 10 days of class.

Family Educational Rights and Privacy Act (FERPA)

Any student who has suppressed their directory information pursuant to Family Educational Rights and Privacy Act (FERPA) should self-identify to the instructor to ensure protection of the privacy of their attendance in this course. See https://registrar.illinois.edu/academic-records/ferpa/ for more information on FERPA.

Sexual Misconduct Policy and Reporting

The University of Illinois is committed to combating sexual misconduct. Faculty and staff members are required to report any instances of sexual misconduct to the University's Title IX and Disability Office. In turn, an individual with the Title IX and Disability Office will provide information about rights and options, including accommodations, support services, the campus disciplinary process, and law enforcement options.

A list of the designated University employees who, as counselors, confidential advisors, and medical professionals, do not have this reporting responsibility and can maintain confidentiality, can be found here: https://wecare.illinois.edu/resources/students/#confidential. Other information about resources and reporting is available at https://wecare.illinois.edu/.

Changes

The instructor reserves the right to make any changes he considers academically advisable. Such changes, if any, will be announced in class and Canvas. Please note that you are responsible for following the class and keeping track of the proceedings.

Tentative Class Schedule

Week	Monday	Wednesday	Friday
1	8/25	8/27	8/29
	Introduction	Neyman-Rubin Causal Model Potential outcome, Causal estimands, Confounding	
2	9/1 Labor day	9/3	9/5
		Randomized experiment Estimation/inference in randomized experiments	
3	9/8	9/10	9/12
	Estimation/inference in randomized experiments	Observational studies Causal assumptions, Identifiability	No class
4	9/15	9/17	9/19
	Matching estimators		Matching estimators (Ruoqi's Lecture)
5	9/22	9/24	9/26
	Inverse propensity score, balancing, outcome regression-based, doubly robust estimators		
6	9/29	10/1	10/3
	IPW, balancing, OR, DR estimators		
7	10/6	10/8	10/10
	Heterogeneous treatment effect Optimal treatment regime		Semiparametric efficiency Double/debiased machine learning (DDML)
8	10/13	10/15	10/17
	Semiparametric efficiency Double/debiased machine learning (DDML)		
9	10/20	10/22	10/24
	Graphical Causal Models DAG, SWIG		Mediation Principal stratum effect, Split treatment
10	10/27	10/29	10/31
	Mediation Principal stratum effect, Split treatment	Instrumental variable	
11	11/3	11/5	11/7
	Instrumental variable	Methods for panal data (DiD, SC, etc.) Regression Discontinuity	
12	11/10	11/12	11/14
	Negative Control Proximal Causal Interference		Interference Dependent Data
13	11/17	11/19 Presentation	11/21 No class
	Interference Dependent Data		
14	11/24	11/26 Thanksgivings	11/28
15	12/1	12/3 Presentation	12/5
16	12/8 Presentation		

Weeks	Topics	Lectures	Assignments
1	Introduction	1	
1	Neyman-Rubin Causal Model	2	
2	Randomized experiment	1	
2,3	Estimation/inference in randomized experiments	2	1
3	Observational studies, Causal assumptions, Identifiability	1	
4	Matching estimators	4.5	1
5,6	IPW, covariate balancing, outcome regression-based, DR estimators	5.5	2
7	Heterogeneous treatment effects Optimal treatment regime	2	1
7,8	Semiparametric efficiency Double/debiased machine learning (DDML)	4	1
9	Graphical Causal Models (DAG, SWIG)	2	
9,10	Mediation, Principal stratum effect, Split treatment	2	1
10,11	Instrumental variable (IV)	3	1
11	DiD/SC/RD	2	1
12	Negative control methods Proximal causal inference	3	1
12,13	Causal inference for dependent data, Interference	2	1
Total		37	11